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Baseline correction using a deep-learning model combining ResNet and UNet

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Most spectral data, such as those obtained *via* infrared, Raman, and mass spectroscopy, have baseline drifts due to fluorescence or other reasons, which have an adverse impact on subsequent analyses. Therefore, several researchers have proposed the use of various baseline-correction methods to address the aforementioned issue. However, most baseline-correction methods require manual adjustment of the parameters to achieve desirable performance. In this study, we propose a baseline-correction method based on a deep-learning model that combines ResNet and UNet. The method uses a deep-learning model trained with simulated spectral data to perform baseline corrections and eliminates the need for manual parameter adjustments. Based on the results of the qualitative and quantitative analyses of the simulated spectral data and actual Raman spectra, the proposed method is easier to apply and has better performance compared to the existing methods. As the proposed method can be applied to Raman spectra and other spectra, it is expected to be widely used.

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1 Introduction

Raman spectroscopy is an effective analytical technique, which can obtain physical and chemical information about trace substances of different materials or living organisms through non-destructive measurements.¹ It is widely used in technical fields, such as food safety, biological research, and hazardous substance detection.^{2–4} However, owing to residual Rayleigh scattering or fluorescence, the measured raw Raman spectra are often disturbed by baseline drifts, which affect subsequent research and analyses. Therefore, baseline correction of spectral data has attracted significant attention.^{5–8}

Baseline and additive noise constitute the main interference in spectral signals. Moreover, the different sources of baseline and additional noise make baseline correction highly difficult. To reduce the interference of a baseline drift, researchers have proposed the use of baseline-correction methods, such as the wavelet transform,^{9–11} polynomial-fitting method,^{12,13} and penalized least squares methods.^{5,14,15}

The baseline-correction method based on the wavelet transform assumes that baseline and additional noises are low-frequency and high-frequency signals in different frequency bands, respectively. Therefore, the baseline of the low-frequency signal can be filtered using a wavelet transform.^{9–11} However, the performance of such a method largely depends on the filtering ability of the wavelet transform. Tuning the

performance of the related methods is difficult because of the use of several parameters such as wavelet selection and scale size.

The polynomial-fitting method fits the baseline by continuously and iteratively eliminating the spectral signal peaks.^{12,13} Although this method is simple and effective, it is prone to overfitting or underfitting and requires artificial observation of each spectrum to determine the appropriate polynomial order for better results.

The penalized least squares method is fast and can avoid peak detection and other user interventions, and is often used for preprocessing baseline correction of various spectral signals. Currently, there are several improved methods based on penalized least squares.^{5,14,15} The rationale behind these methods is to iteratively change weights by estimating a baseline. If the spectral signal is lower than the previously fitted baseline, it is assigned a larger weight; otherwise, it is assigned a lower weight or is not weighted. Changes in the weights are adjusted by setting the parameters manually based on the input spectrum. Therefore, if the parameters are fixed, the baseline-correction result varies significantly based on the input spectrum.

Deep learning is a key field of artificial intelligence and plays a crucial role in the areas of computer vision, pattern recognition, signal processing, and other technological fields. However, few researchers have used deep-learning techniques for baseline correction of spectral signals. This is because deep learning requires numerous labeled datasets to train a model. Acquiring manually labeled spectral signals that are sufficient for training deep-learning networks is difficult.

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Moreover, the cost of collecting and labeling large amounts of spectral data is prohibitive because spectral data cannot be obtained for free. Therefore, Kung *et al.*¹⁶ and Liu¹⁷ generated simulated spectral data and then utilized them for deep learning. For baseline correction, the widely used ResNet model is used and better results are obtained compared to the existing methods.

Herein, we use a data-generation method and a new deep-learning network combining ResNet (residual network)¹⁸ and UNet (U-shaped network)¹⁹ to improve the performance of the existing baseline-correction methods. The subsequent section briefly reviews the existing methods, followed by a description of the proposed deep-learning network. In the following section, the proposed method is compared with the existing methods in terms of experimental results obtained using simulation data, and the baseline-correction results of the actual Raman data are presented.

2 Previous methods

2.1 Penalized least squares methods

Penalized least squares regression provides a method for balancing data fitting closely and avoiding excessive roughness or rapid variation. Therefore, it can be used to estimate the baseline for a given spectrum. The penalized least squares method and its variants have shown good performance in spectral baseline correction, and several studies have been performed on this method in recent years.

Suppose that $\mathbf{y}$ is a spectral signal of length N sampled at equal intervals and $\mathbf{z}$ is the fitted spectral signal to be identified. Assuming that $\mathbf{D}$ is the difference matrix, $\mathbf{z}$ can be obtained by minimizing the following regularized least-squares function.

$$S(\mathbf{z}) = (\mathbf{y} - \mathbf{z})^T(\mathbf{y} - \mathbf{z}) + \lambda(\mathbf{D}\mathbf{z})^T\mathbf{D}\mathbf{z}. \quad (1)$$

The first and second terms in eqn (1) represent the fitness of the data and the smoothness of $\mathbf{z}$, respectively. The parameter λ in eqn (1) can adjust the balance between these two terms. In this method, a weight vector $\mathbf{W}$ can be introduced to estimate the baseline by applying a penalty based on the spectrum signal. Assuming that $\mathbf{W}$ is a diagonal matrix with the element w , eqn (1) can be changed to the following penalized least squares function:

$$S(\mathbf{z}) = (\mathbf{y} - \mathbf{z})^T\mathbf{W}(\mathbf{y} - \mathbf{z}) + \lambda\mathbf{z}^T\mathbf{D}^T\mathbf{D}\mathbf{z}. \quad (2)$$

By setting the partial derivatives of eqn (2) to 0, the solution to the minimization problem can be expressed as follows:

$$\mathbf{z} = (\mathbf{W} + \lambda\mathbf{D}^T\mathbf{D})^{-1}\mathbf{W}\mathbf{y}. \quad (3)$$

If this method is used for baseline correction, the peak regions must be known in advance; the weight w_i of these regions was set to 0 and w_i of the non-peak regions was set to 1. To avoid peak detection, the asymmetric least squares (AsLS)²⁰ and adaptive iteratively reweighted least

squares (airPLS)⁵ methods have been proposed as iterative methods to obtain the weights $\mathbf{w}$ adaptively. The two methods are similar; however, airPLS adopts a different weight-allocation method and adds a penalty to control the smoothness of the fitted baseline.

The AsLS and airPLS methods have a common disadvantage—a boosted baseline-corrected spectrum is obtained when handling the spectrum in additive noise. In this respect, the asymmetrically reweighted penalized least squares (arPLS)¹⁴ method introduced asymmetrical weights to prevent the estimated baseline from being boosted. Although this method can ensure a satisfactory baseline in the peak-free region, it is prone to misjudgment in the peak region. This is because the tails of the peak are considerably low and are often misjudged as non-peak regions.

The adaptive smoothness parameter penalized least squares (asPLS)¹⁵ method can solve the problem associated with the method mentioned above. The main idea of asPLS is to set the parameter λ to a large value in the peak regions and to a small value in the non-peak regions by introducing a coefficient vector α that can be obtained using eqn (4).

$$\alpha_i = \frac{\text{abs}(y_i - z_i)}{\max(\text{abs}(\mathbf{y} - \mathbf{z}))}. \quad (4)$$

$$\mathbf{z} = (\mathbf{W} + \lambda\alpha\mathbf{D}^T\mathbf{D})^{-1}\mathbf{W}\mathbf{y}. \quad (5)$$

From eqn (4), it is evident that α_i is set to a larger value in the peak position regions and *vice versa*. Now, eqn (5) replaces eqn (3) in asPLS.

2.2 Deep-learning methods

In recent years, deep-learning technology has played an important role in several research fields. However, the deep-learning technology is rarely used for spectral baseline correction owing to the difficulty in obtaining a large amount of training data. To overcome the problem of small training data and use deep-learning techniques for spectral baseline correction, Kung *et al.* and Liu generated simulated spectral data and used them to train deep-learning models. Both applied convolutional neural networks (CNNs) for deep-learning networks for baseline correction and adopted ResNet blocks to improve the performance of conventional CNNs.

The more layers a deep-learning network has, the better it can extract representative features from the training data, resulting in better training results. However, when a convolutional neural network becomes extremely deep, the gradient values saturate with the values that are considerably large or small because of the vanishing-gradient problem and do not move during parameter updates. Therefore, the learning effect disappears or the learning speed is significantly slow.

ResNet was proposed to solve the vanishing-gradient problem through the shortcut connection of the residual learning framework. Fig. 1 shows the structure of the ResNet block; Fig. 1a is an identity shortcut and Fig. 1b is a projection shortcut. The projection shortcut adjusts the outputs of the front

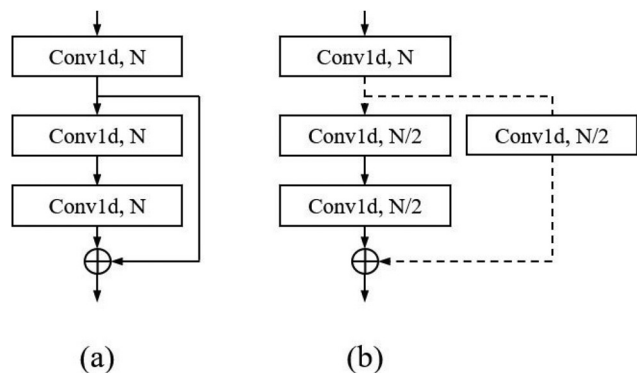


Fig. 1 Structure of the ResNet block. (a) Identity shortcut. (b) Projection shortcut.

and rear ResNet blocks by downsampling such that they can be added to each other.

Because of the advantages of ResNet, the two baseline-correction studies mentioned above used it to construct their deep-learning model for the experiments. The results of their studies demonstrated that the deep-learning-based baseline-correction methods are better than the traditional methods in this regard. Moreover, it was unnecessary to manually adjust the parameters when applying the model, and the number of layers or channels had to be determined when creating the model.

3 Proposed method

3.1 Data-generation method

The success of deep-learning models depends on the amount and variety of training data, the structure of the deep-learning network, and hyperparameter tuning. Because it is difficult to obtain sufficient spectral data for the successful training of deep-learning networks, there have been few studies on the baseline correction of spectral data based on deep-learning techniques. In this study, we use our method to generate a simulated training dataset.

Fig. 2 shows the data-generation procedure used in this experiment. The baseline and peaks of the spectral data were generated and combined to form a simulated spectrum.

First, the number and positions of the peaks were randomly determined such that various peaks can be expressed. This number was set between 5 and 15. The height and width of the peaks were then randomly selected within a certain range. In the experiments, we used 0.025 to 1 and 5 to 21. These positions, heights, and widths were used to model the peaks using a Hanning window. The baseline was modeled using anchor points and interpolation. The number of anchor points was set between 2 and 7. A baseline of length 512 was generated through cubic spline interpolation using these anchor points.

The final simulated spectrum was obtained by combining the baseline, peak, and additional Gaussian white noise. When mixing the baseline and the peaks, we set the mixing

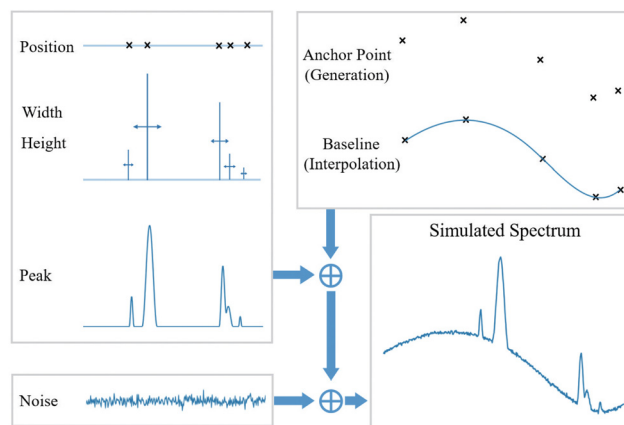


Fig. 2 Procedure for generating simulation data.

ratio to be between 0.1 and 0.8. In addition, the baseline and peaks were combined using random flips to ensure data diversity. The simulated spectra were normalized to relative intensities ranging from 0 to 1 using a min-max normalization method to minimize deviations due to differences in relative intensities. Fig. 3 shows the examples of the simulated data generated using this procedure. The figure confirms that the spectral data simulated using the proposed method can depict various baselines and peaks.

3.2 Proposed deep-learning model

The convolutional layer has fewer parameters and retains the spatial information of the input data compared to the fully connected layer.

As previously described, residual learning can extract more representative features from the input data by solving the vanishing-gradient problem during deep-learning network train-

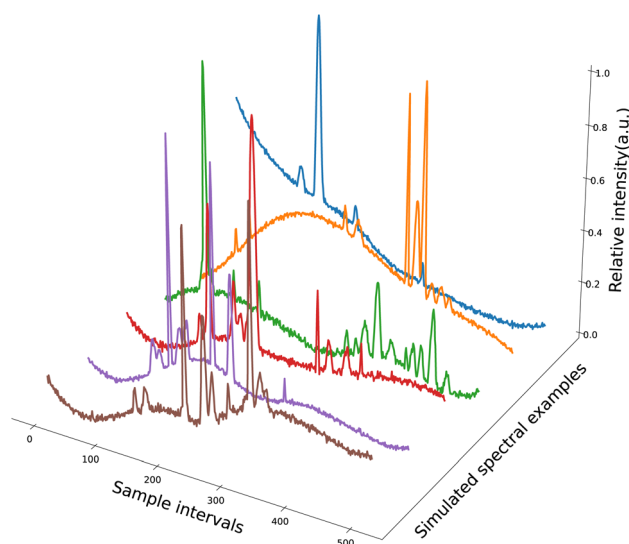


Fig. 3 Simulated data examples with different baseline and additional noise.

ing. ResNet has a better regression and recognition ability compared to the conventional CNN. Therefore, we adopted convolutional layers with ResNet.

UNet was originally applied in the field of medical images and named after its U-shaped network structure. It comprises contracting and expansive paths. The contracting path captures the context from the input data, as observed in the general deep-learning structure. UNet's main idea is to upsample the feature map on the expansive path and combine the context of the feature map captured on the contracting path for more accurate localization. An advantage of UNet is that it can retrieve the lost pixel information because the size of the data changes from small to large when passing through the contracting and expansive paths by skipping the connection.

In this study, we construct a deep-learning model based on CNN by combining UNet with ResNet to enhance the performance of the baseline-correction method. The main structure of the proposed deep-learning network, ResUNet (UNet with ResNet), is similar to that of UNet; however, ResNet is applied to the expansion and contracting paths.

Furthermore, we used a one-dimensional convolutional layer (conv1d) for downsampling instead of Maxpooling such that the output channel size of each layer can be freely set to the desired size. In upsampling, a one-dimensional transposed convolutional layer (ConvTranspose1d) was used to maintain consistency with downsampling. The final output of the proposed model uses a convolutional layer, instead of a fully connected linear layer to reduce the computational cost of the model.

Fig. 4 shows the structure of ResUNet. Table 1 shows the architectural table of ResUNet. The table also summarizes the output size of all the tiers and the network-architecture parameters, such as tier numbers and parameters.

4 Experiments and results

The ResUNet model is compared to existing methods to evaluate its performance. Among the penalized least-squares-based methods, arPLS and asPLS are selected, and CNN with ResNet and CNN with UNet are selected for the deep-learning method.

4.1 Preliminary experiments

Before the formal experiment, we used a small amount of data for the preliminary experiments to determine the best parameters and reduce experimental time. The preliminary experiments were mainly conducted in two aspects: selecting the best smoothness parameter λ of arPLS and asPLS and determining the channel size of the deep-learning-based methods.

If the smoothness parameter λ of arPLS and asPLS is considerably large or small, it affects the performance of the baseline correction and cannot obtain satisfactory results. Therefore, before conducting the main experiments, a proper smoothness parameter λ should be determined. We used the root mean square error (RMSE) to evaluate the baseline-correction performance. Assuming that the true and fitted baselines are y and z , respectively, RMSE is defined as follows.

$$\text{RMSE} = \left[\sum_{i=1}^N (y_i - z_i)^2 / N \right]^{\frac{1}{2}} \quad (6)$$

The value of λ ranging from 10^0 to 10^5 was selected on a log scale to determine the optimal smoothing parameter.²⁰ Table 2 lists the RMSE values of baseline correction for different values of the smoothness parameter λ from the two methods using 20,000 training data. From the table, it can be concluded that when the value of the parameter λ of the arPLS method is 10^1 and the value of the parameter λ of the asPLS

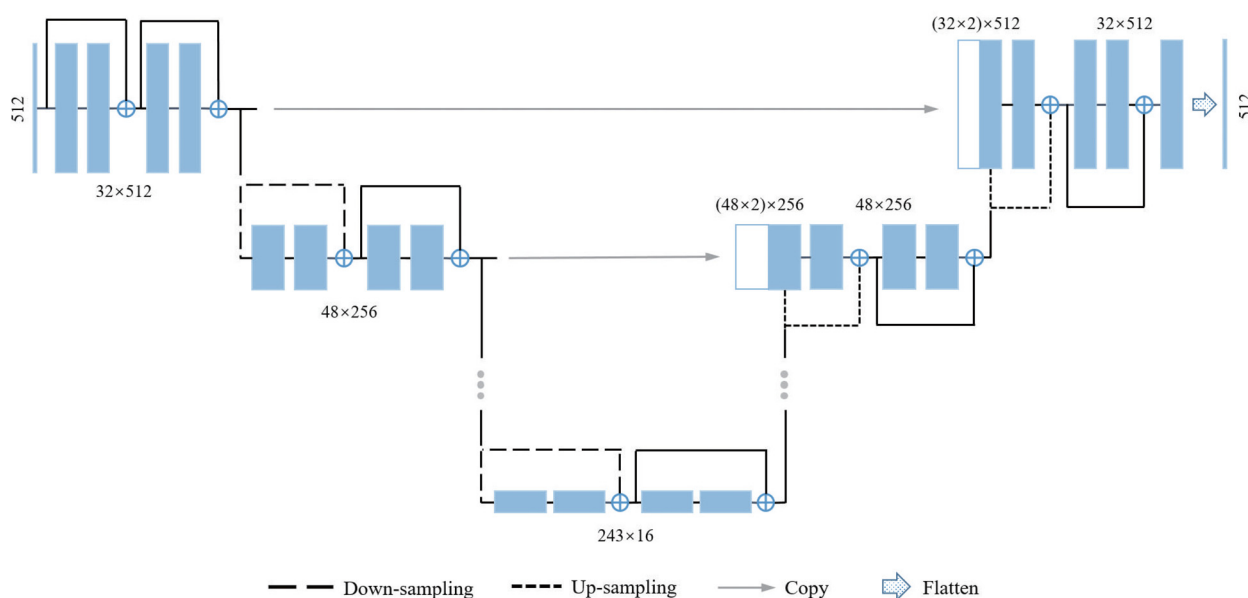


Fig. 4 Structure of the proposed ResUNet.

Table 1 Architecture of the proposed ResUNet model

	Layer	Repeat	Output	Parameter
Block ×4	Conv1d	1	32 × 512	128
	Conv1d	4	32 × 512	12 416
	Conv1d	1	48 × 256	592 098
	Conv1d	3	72 × 128	
Block ×1	Conv1d (ShortCut)	1	108 × 64	
			162 × 32	
	Conv1d	1	243 × 16	768 852
	Conv1d	3		
Block ×5	Conv1d (ShortCut)	1		
	ConvTranspose1d	1	162 × 32	976 930
	Conv1d	3	108 × 64	
	ConvTranspose1d (ShortCut)	1	72 × 128	
			48 × 256	
			32 × 512	
	Conv1d	1	1 × 512	33
	Flatten	1	512	0
Total number of parameters				2 350 457

Table 2 RMSE ($\times 10^{-4}$) of baseline-corrected spectra for different values of parameter λ

	$\log_{10} \lambda$					
	0	1	2	3	4	5
arPLS	243.74	82.25	128.13	577.62	1349.30	1923.00
asPLS	825.87	615.52	242.62	93.21	63.90	337.05

method is 10^4 , the RMSE value is the smallest; that is, the performance of the two methods is best for the preliminary data.

In the case of two-dimensional data in deep learning, the channel size doubles when the number of layers increases. In this study, an experiment was conducted by setting the number of channels using one-dimensional data, as shown in eqn (7).

$$C_{i+1} = \beta C_i, 1 \leq \beta \leq 2. \quad (7)$$

In this study, the first channel size C_1 was set to 16 and 32, and β was set to 1.5 and 2. The preliminary experiments to determine the channel size were conducted using ResNet. 60 000 simulated spectra were used to train ResNet with different channel sizes and 20 000 test spectra were used to verify the performance of ResNet using RMSE.

Based on the preliminary experimental results of the ResNet shown in Table 3, it can be said that it is better to set the channel size to an appropriate number rather than down-sampling or up-sampling by a multiple of 2. Evidently, the appropriate channel size can be adjusted to reduce the number of parameters. Using the same data, the results of UNet and ResUNet are shown for reference. In the main experiments, β was set to be 1.5 based on the aforementioned results.

4.2 Training strategy

In this study, the training loss function uses RMSE and the validation loss function uses the mean absolute error (MAE). The adaptive moment estimation (Adam) algorithm was adopted as the optimization method. The batch size and maximum learning epoch of the deep-learning model were set to be 500 and 1200, respectively.

The learning rate is set to 5×10^{-4} and a decreasing setting is adopted. If the minimum value of the validation loss does not change for more than 75 epochs, it is reduced to 4/5 of the starting learning rate; if it is reduced to 1/8 of the original learning rate, the model stops training. Finally, if the current validation loss is less than that of the previous one, the current model is saved.

Table 3 Results of the preliminary experiments for constructing a deep-learning network

Model	Channel						Parameter	RMSE ($\times 10^{-4}$)
ResNet	16	32	64	128	256	512	6 427 360	6.66
	16	32	64	128	256		3 180 640	7.15
	32	64	128	256	512		8 521 408	7.10
	32	48	72	108	162	243	2 394 762	6.66
UNet	32	48	72	108	162	243	2 070 882	6.20
ResUNet	32	48	72	108	162	243	2 350 457	5.85

In this study, all the deep-learning models were trained on a computer with a GeForce RTX 3080 GPU. Pytorch version 1.7.0 was used as the deep-learning framework.

4.3 Simulated data

In the experiments, 210 000 simulated spectra were generated for training, whereas 70 000 simulated spectra were used for testing. The proposed method was compared to two penalized least-squares-based methods, arPLS and asPLS, and two deep-learning-based methods, ResNet and UNet. We compared the baseline-correction performance in terms of the RMSE and MAE, and the results are summarized in Table 4 and Fig. 5.

$$\text{MAE} = \sum_{i=1}^N |y_i - z_i| / N \quad (8)$$

Table 4 shows that the deep-learning-based baseline-correction method is significantly better than the two traditional methods, arPLS and asPLS. This also demonstrates that the proposed method is superior to the ResNet and UNet models.

When baseline correction was accurately performed, the results were almost the same, particularly with the deep-learning models. Therefore, it is expedient to observe the performance of these models when the baseline is relatively poorly fitted. To examine this, the standard deviations of the 10 000, 1000, and 500 spectra with poor results from 70 000 were calculated. From Table 5, it can be observed that the standard deviation of the ResUNet model is the smallest. In particular, the poorer the baseline-correction result, the smaller the deviation. This indicates that the larger the baseline-correction error, the better the results of the proposed method.

To observe the qualitative performance of each method, two results with a low fit were selected as shown in Fig. 6 and 7. Evidently, the other methods overestimated or underestimated the baseline of the spectrum in the magnified position compared to the proposed method.

Table 4 Error values ($\times 10^{-4}$) of baseline-corrected spectra obtained using various methods

	Baseline-correction methods				
	arPLS	asPLS	ResNet	UNet	ResUNet
RMSE	59.75	39.64	6.11	5.68	5.40
MAE	26.81	21.27	4.77	4.43	4.24

Table 5 Standard deviation ($\times 10^{-4}$) of RMSE and MAE obtained using deep-learning methods

Metric	Amount	ResNet	Net	ResUNet
RMSE	10 000	3.09	2.87	2.74
	1000	4.42	3.80	3.40
	500	5.30	4.47	3.98
MAE	10 000	2.06	2.00	1.97
	1000	2.01	1.81	1.72
	500	2.20	1.92	1.79

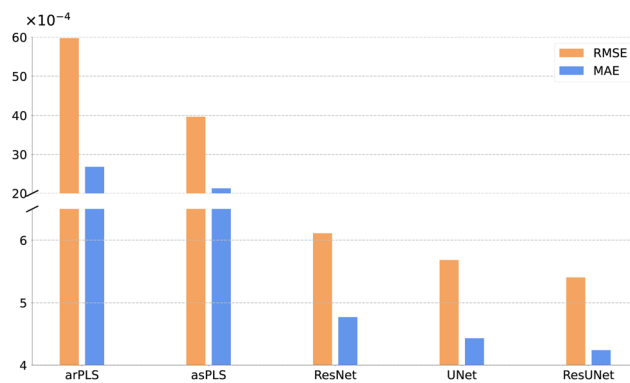


Fig. 5 RMSE and MAE of baseline-corrected spectra obtained by various methods.

Notwithstanding the quantitative analyses in Tables 4 and 5 or the qualitative analyses in Fig. 6 and 7, the deep-learning-based baseline-correction methods had outstanding performance compared to the penalized least-squares-based baseline-correction methods. The performance of the proposed method was better than those of the other two deep-learning methods.

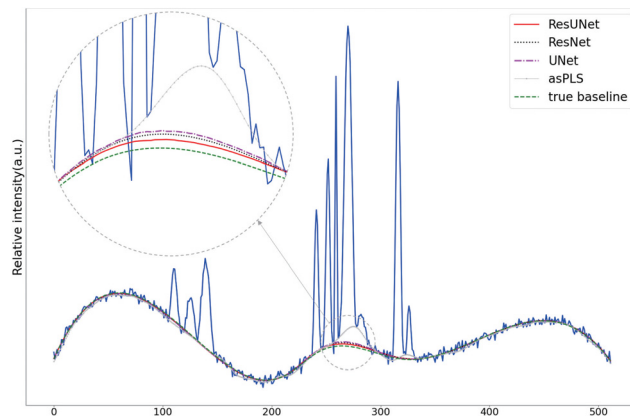


Fig. 6 Baseline-fitting results of the first simulation spectrum.

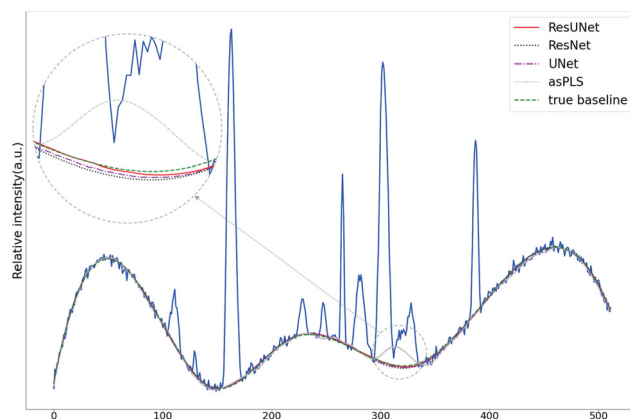


Fig. 7 Baseline-fitting results of the second simulation spectrum.

4.4 Experimental Raman spectrum

In this study, the Raman spectra of 35DNT (3,5-dinitrotoluene) and 4ADNT (4-amino-2,6-dinitrotoluene) were selected for the experiments. When measuring the Raman spectra, the laser power was maintained at 1.0 mW or less to prevent the effect of laser heating.²¹ A holographic notch filter was used in the measurement process to remove the Rayleigh line from Raman scattering. The Raman spectroscopic signal was accumulated for 5 s and the laser was focused using a 50× objective lens.

The Raman spectra of the above two substances were selected to demonstrate the performance of the proposed deep-learning model. The 35DNT and 4ADNT spectra were selected as the examples of linear and highly curved backgrounds, respectively. To apply these spectra to the deep-learning model, the length of these spectra was set to 512 and the relative intensity was set to 1 through preprocessing. Fig. 8 and 10 show two sets of 50 spectra for these substances. As shown in these figures, although the spectra were obtained by measuring the same substance, they showed different baselines.

The baseline-corrected spectra obtained using the proposed deep-learning model are shown in Fig. 9 and 11. Evidently, the

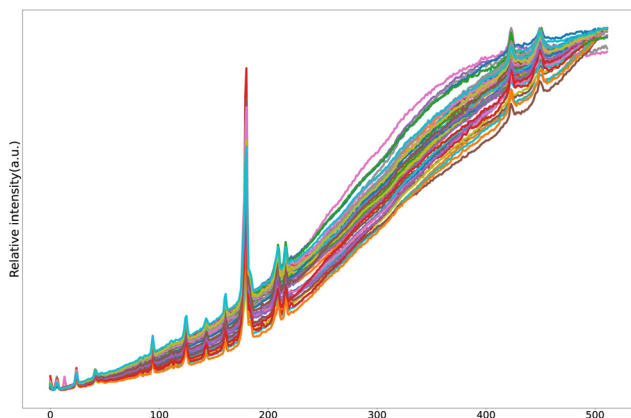


Fig. 8 Measured 35DNT Raman spectra.

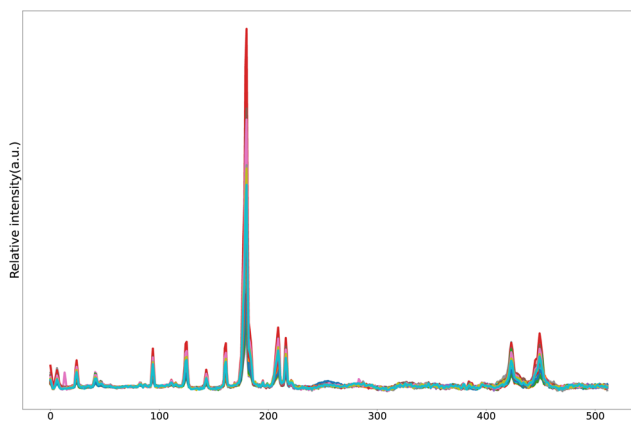


Fig. 9 Baseline-corrected 35DNT Raman spectra.

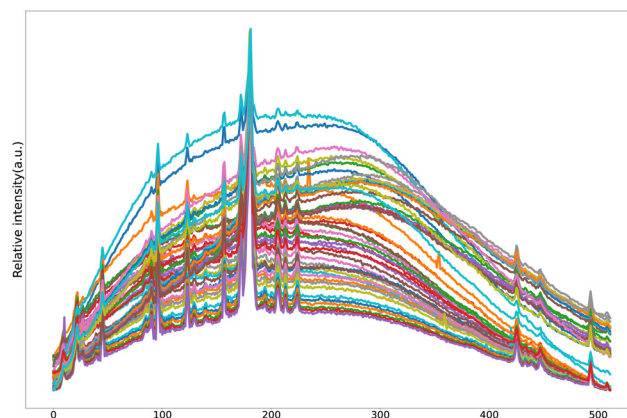


Fig. 10 Measured 4ADNT Raman spectra.

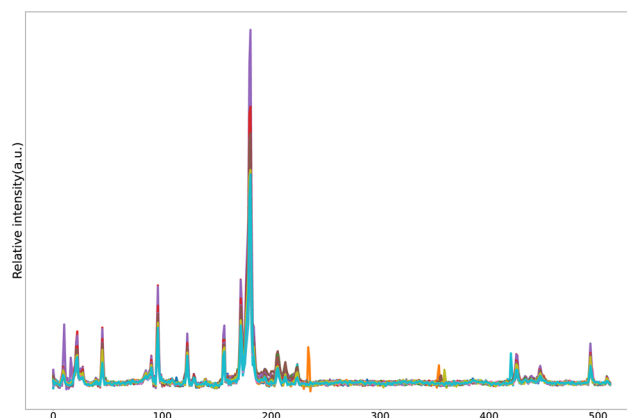


Fig. 11 Baseline-corrected 4ADNT Raman spectra.

baseline-corrected spectra are natural and desirable. Therefore, the proposed method can successfully correct the baseline of the Raman spectra of the two substances. Moreover, the proposed method can be applied to a variety of Raman spectra, and the results will be satisfactory.

5 Conclusions

In this study, a baseline-correction method for spectra was proposed based on a deep-learning model comprising a combination of UNet and ResNet. Compared to existing penalized least squares methods, such as arPLS and asPLS, the proposed method does not require subjective parameter estimation and shows better performance.

Two problems were addressed when applying the proposed deep-learning technique to baseline correction. First, a method for simulating spectra by randomly generating baseline, peak, and additive noise and combining them was proposed to address the problem of insufficient training data. In addition, it was demonstrated that, for one-dimensional data, using an appropriate channel size rather than doubling the

channels of the previous layer can reduce the number of model parameters and improve the performance of the model.

Through comparative experiments with existing deep-learning-based methods using simulated data, it was found that the proposed method can achieve excellent accuracy without increasing the number of parameters. Moreover, the standard deviation of the results obtained using the proposed model was the smallest amongst all the methods that were considered for comparison. We hope that the proposed method can be widely used for baseline correction of the measured spectra.

Conflicts of interest

There are no conflicts to declare.

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